

1.a Find  $\frac{\partial f}{\partial x}(1,0)$  where  $f(x) = x^3 + y^3 + xy$ .

Solution.

$$\begin{aligned} f(x) &= x^3 + y^3 + xy \text{ finding } \frac{\partial f}{\partial x} \\ \text{partial derivative in respect to } x & \\ &= 3x^2 + y \text{ where } (1,0) \\ &= 3(1)^2 + 0 = \underline{3}. \end{aligned}$$

1b Find the total derivative  $\frac{dz}{dt}$  of  $z = 7u + vt$ , where  $u = 2t^2$  and  $v = t + 1$ .

Solution.

$$\begin{aligned} z &= 7u + vt, \quad u = 2t^2, \quad v = t + 1 \\ \text{Subst } u \text{ and } v \text{ in } z &= 7(2t^2) + (t+1)t = 14t^2 + t^2 + t \\ \text{finding } \frac{dz}{dt} &= 28t + 2t + 1 \\ &= \underline{30t + 1}. \end{aligned}$$

2a Given the linear equation  $\frac{dy}{dx} - 2y = 3e^x$ , whose solution passes through the origin. Employ Taylor's series to find  $y$  at  $x = 0.1$  to 4dp.

Solution.

$$\frac{dy}{dx} - 2y = 3e^x \therefore \frac{dy}{dx} = 3e^x + 2y$$

~~Taylor's series:~~ 
$$f(y(x)) = f(x_0, y_0) + (x-x_0)f_x(x_0, y_0) + (y-y_0)f_y(x_0, y_0) + \frac{(x-x_0)^2}{2}f_{xx}(x_0, y_0) + \frac{(y-y_0)^2}{2}f_{yy}(x_0, y_0) + 2 \int$$

$$\text{Taylor series} = f(x_0) + (x-x_0)f'(x_0) + \frac{(x-x_0)^2}{2!}f''(x_0) + \frac{(x-x_0)^3}{3!}f'''(x_0)$$

Given that the solution passes the origin  $\therefore y = 0$

$$f'(x) = 2y + 3e^x \text{ where } y = 0, x = 0 = 2(0) + 3e^0 = 0 + 3 = \underline{3}.$$

$$f''(x) = 2\frac{dy}{dx} + 3e^x \text{ where } \frac{dy}{dx} = 3 \therefore 2(3) + 3e^0 = 6 + 3 = 9.$$

$$f'''(x) = 2\frac{d^2y}{dx^2} + 3e^x \text{ where } \frac{d^2y}{dx^2} = 9 \therefore 2(9) + 3e^0 = 18 + 3 = 21.$$

$$f^{(4)}(x) = 2\frac{d^3y}{dx^3} + 3e^x \text{ where } \frac{d^3y}{dx^3} = 21 \therefore 2(21) + 3e^0 = 42 + 3 = 45.$$

Taylor series expansion at  $x=0$

$$f(x) = y(0) + \frac{y'(0)}{1!}x + \frac{y''(0)}{2!}x^2 + \frac{y'''(0)}{3!}x^3 + \frac{y^{(4)}(0)}{4!}x^4$$

Substituting.

$$y(x) = 0 + \frac{3}{1!}x + \frac{9}{2!}x^2 + \frac{21}{3!}x^3 + \frac{45}{4!}x^4$$

$$y(x) = 0 + 3x + 4.5x^2 + \frac{21}{6}x^3 + \frac{45}{24}x^4 = 3x + 4.5x^2 + 3.5x^3 + 1.875x^4$$

Substituting  $x=0.1$ .

$$y(0.1) = 3(0.1) + 4.5(0.1)^2 + 3.5(0.1)^3 + 1.875(0.1)^4 \\ = 0.3 + 0.045 + 0.0035 + 0.001875 = \underline{0.3504} \text{ to 4dp.}$$

2b Solve  $dy = (xy-1)dx$  given  $y(1)=2$  using Taylor series

solution

$dy = (xy-1)dx$  divide both sides by  $dx$ ,  $\frac{dy}{dx} = (xy-1)$  find  $x$  at 1.02.

where  $y(x) = y(1) = 2$  where  $x=1$  and  $y=2$ .

$$f'(x) = xy-1 \therefore (1)(2)-1 = 2-1 = 1.$$

$$f''(x) = \frac{dy}{dx}x + y \therefore 1(1) + 2 = 3.$$

$$f'''(x) = \frac{d^2y}{dx^2}x + 2\frac{dy}{dx} = 3(1) + 2(1) = 5$$

$$f^{(4)}(x) = \frac{d^3y}{dx^3}x + 3\frac{d^2y}{dx^2} = 5(1) + 3(3) = 5+9 = 14.$$

Taylor's series expansion at  $x=1$ .

$$y(x) = y(1) + \frac{y'(1)}{1!}(x-1) + \frac{y''(1)}{2!}(x-1)^2 + \frac{y'''(1)}{3!}(x-1)^3 + \frac{y^{(4)}(1)}{4!}(x-1)^4$$

Substituting where  $y=2$   $x=1.02$  and  $(x-1) = 1.02-1 = 0.02$ .

$$y(1.02) = 2(1) + \frac{1}{1!}(0.02) + \frac{3}{2!}(0.02)^2 + \frac{5}{3!}(0.02)^3 + \frac{14}{4!}(0.02)^4 \\ = 2 + 0.02 + \frac{3}{2}(0.0004) + \frac{5}{6}(0.000008) + \frac{14}{24}(0.00000016)$$



$$= 2 + 0.02 + 0.0006 + 0.00000667 + 0.0000000933 = 2.0206$$

$$\therefore y(1.02) = 2.0206 \text{ to 5dp.}$$

3a Use NR method to find real root of  $x^2 - 2x - 5 = 0$  to 3dp.  
Solution.

$$\text{NR: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad \text{Finding } x_0 \text{ where } f(x_0) \text{ is closer to 0.}$$

$$f(1) = 1^2 - 2(1) - 5 = -6$$

$$\therefore x_0 = 3 \quad f' = 2x - 2$$

$$f(2) = 2^2 - 2(2) - 5 = -5$$

$$\text{finding } x_1 = x_0$$

$$f(3) = 3^2 - 2(3) - 5 = -2 \rightarrow \text{closer to 0.}$$

$$f(4) = 4^2 - 2(4) - 5 = 3$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = 3 - \frac{(3^2 - 2(3) - 5)}{(2(3) - 2)} = 3 - \frac{(-2)}{4} = 3 + \frac{2}{4} = 3.5$$

$$\therefore x_1 \neq x_0$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 3.5 - \frac{(3.5^2 - 2(3.5) - 5)}{2(3.5) - 2} = 3.45 \quad \therefore x_2 \neq x_1$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 3.45 - \frac{(3.45^2 - 2(3.45) - 5)}{2(3.45) - 2} = 3.4495 \quad \therefore x_3 \neq x_2$$

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 3.4495 - \frac{(3.4495^2 - 2(3.4495) - 5)}{2(3.4495) - 2} = 3.4495 \quad x_4 = x_3$$

$$x_4 = x_3 = 3.4495$$

3b Carry out the iterations finding real root of  $x \sin x + \cos x = 0$  near  $x = \pi$ .

$$\text{Solution: } x = \pi \quad \text{where NR: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f'(x) = \frac{d}{dx}(x \sin x) + \frac{d}{dx}(\cos x)$$

$$\text{taking } x \sin x \text{ using product rule } u \frac{dv}{dx} + v \frac{du}{dx} \quad \frac{dv}{dx} = \cos x \quad \frac{du}{dx} = 1 \therefore \downarrow$$

$$= x(\cos x) + \sin x(1)$$

$$\text{taking } \cos x \quad f(x)' = -\sin x \quad \therefore f'(x) = (x(\cos x) + \sin x) + (-\sin x)$$

$$= x \cos x$$

Iterating NR where  $x_0 = \pi$

$$x_1 = \pi - \frac{f(x_0)}{f'(x_0)} = \pi - \frac{(\pi \sin \pi + \cos \pi)}{(\pi \cos \pi)} \quad \text{where } \begin{matrix} \pi = \pi \\ \sin \pi = 0 \\ \cos \pi = -1 \end{matrix}$$

$$\therefore x_1 = \pi - \frac{(\pi(0) + (-1))}{\pi(-1)} = \pi - \left(\frac{-1}{-1}\right) = \pi - \frac{1}{\pi}$$

$$\text{for } x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \quad \text{where } x_1 = \pi - \frac{1}{\pi}$$

$$x_2 = \pi - \frac{1}{\pi} - \frac{\left(\pi - \frac{1}{\pi}\right) \sin\left(\pi - \frac{1}{\pi}\right) + \cos\left(\pi - \frac{1}{\pi}\right)}{\left(\pi - \frac{1}{\pi}\right) \cos\left(\pi - \frac{1}{\pi}\right)}$$

$$\text{where } \sin\left(\pi - \frac{1}{\pi}\right) = -\sin\left(\frac{1}{\pi}\right)$$

$$\cos\left(\pi - \frac{1}{\pi}\right) = -\cos\left(\frac{1}{\pi}\right)$$

$$\therefore x_2 = \frac{\left(\pi - \frac{1}{\pi}\right) \left(-\sin\left(\frac{1}{\pi}\right)\right) + \left(-\cos\left(\frac{1}{\pi}\right)\right)}{\left(\pi - \frac{1}{\pi}\right) \left(-\cos\left(\frac{1}{\pi}\right)\right)}$$

$$\therefore x_2 = \underline{x_1}$$

4a If  $y = 4x^3 - 5x^2 + 6$ ,  $\frac{d^2y}{dx^2}$

Solution

$$y = 4x^3 - 5x^2 + 6 \quad \frac{dy}{dx} = 12x^2 - 10x \quad \frac{d^2y}{dx^2} = 24x - 10$$

4b If  $y = 3x^3 - 2x + 1$  and  $\frac{d^2y}{dx^2} = 4$ , find  $x$

Solution

$$y = 3x^3 - 2x + 1 \quad \frac{dy}{dx} = 9x^2 - 2 \quad \frac{d^2y}{dx^2} = 18x \quad \text{where } \frac{d^2y}{dx^2} = 4$$
$$= \frac{18x}{18} = \frac{4}{18} \quad x = \frac{2}{9}$$



59 Find the point on the curve  $y = x^2 - 5x + 6$ , and the area cut off by  $x$  axis.

Solution.

$$y = x^2 - 5x + 6 = (x^2 - 3x)(-2x + 6) = x(x-3) - 2(x-3)$$

$x = 3$  and  $x = 2$  points on curve = 2, 3.

Area cut off =  $\int_2^3 (x^2 - 5x + 6) dx$

$$= \frac{x^{2+1}}{2+1} - \frac{5x^{1+1}}{1+1} + \frac{6x^{0+1}}{0+1} \Big|_2^3 = \frac{x^3}{3} - \frac{5x^2}{2} + 6x \Big|_2^3$$

$$\therefore \left[ \frac{3^3}{3} - \frac{5(3)^2}{2} + 6(3) \right] - \left[ \frac{2^3}{3} - \frac{5(2)^2}{2} + 6(2) \right]$$

$$= \left[ \frac{27}{3} - \frac{45}{2} + 18 \right] - \left[ \frac{8}{3} - \frac{20}{2} + 12 \right] = -\frac{1}{6} \text{ unit square.}$$

56 Given that  $x^2 + y^2 = 25$  - find  $\frac{d^2y}{dx^2}$  at  $p(4, 3)$ .

Solution

$$x^2 + y^2 = 25 \quad dy/dx = 2x + 2y \frac{dy}{dx} = 0 \quad \frac{dy}{dx} = \frac{-2x}{2y} = \frac{-x}{y}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left( \frac{-x}{y} \right) \quad \text{let } -x \text{ be } u \text{ and } y \text{ be } v \text{ using quotient rule}$$
$$= \frac{\frac{du}{dx} \cdot v - u \cdot \frac{dv}{dx}}{v^2} = \frac{\frac{du}{dx} \cdot v - u \cdot \frac{dv}{dx}}{v^2}$$

$$\therefore \frac{d^2y}{dx^2} = \frac{y(-1) - (-x)\left(\frac{dy}{dx}\right)}{y^2} \quad \text{where } dy/dx = \frac{-x}{y}$$

$$\therefore \frac{-y - (-x)\left(\frac{-x}{y}\right)}{y^2} = \frac{-y - \frac{x^2}{y}}{y^2} \quad \text{at } p(4, 3)$$

$$\therefore \frac{-3 - \frac{(4)^2}{3}}{(3)^2} = \frac{-3 - \frac{16}{3}}{9} = \frac{-\frac{25}{3}}{9} = \frac{-25}{3} \times \frac{1}{9} = \frac{-25}{27}$$

6a Verify the Mean value theorem for  $f(x) = x^2 - 4x - 3$  on  $[1, 4]$ .

Solution

$$f(x) = x^2 - 4x - 3 \text{ at } [1, 4]$$

$$f'(x) = 2x - 4$$

$$f'(a) = 2(1) - 4 = -2 \quad f(a) = 1^2 - 4(1) - 3 = -6 \quad a = 1 \quad f'(b) = 2(4) - 4 = 4$$

$$f(b) = 4^2 - 4(4) - 3 = -3 \quad b = 4$$

$$\text{Mean value theorem} = f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\therefore 2c - 4 = \frac{-3 - (-6)}{4 - 1} = \frac{3}{3} = 1$$

$$2c - 4 = 1, \quad 2c = 5$$

$$c = \frac{5}{2}$$

6b Mean value theorem of  $f(x) = x^3$  on  $[-1, 3]$

Solution

$$f(x) = x^3$$

$$f'(x) = 3x^2$$

$$f(a) = (-1)^3 = -1$$

$$f(b) = (3)^3 = 27$$

$$\therefore 3c^2 = \frac{27 - (-1)}{3 - (-1)} = \frac{28}{4} = 7$$

$$3c^2 = 7 \quad c^2 = \frac{7}{3} \quad c = \sqrt{\frac{7}{3}}$$

7a Find the partial derivative of  $xy^2 + 3x^2e^y$

Solution

$$xy^2 + 3x^2e^y = f(x, y)$$

Solving partially: with respect to  $x$

$$\therefore \frac{\partial f}{\partial x} = y^2 + 6xe^y$$

Solving partially with respect to  $y$

$$\therefore \frac{\partial f}{\partial y} = 2xy + 3x^2e^y$$



7b Find the partial derivative of  ~~$x^2$~~   $2x \sin y + y^2 \cos x$ .

Solution

$$f(x, y) = 2x \sin y + y^2 \cos x$$

partially with respect to  $x$ :

$$\frac{\partial f}{\partial x} = 2 \sin y + \cancel{y^2} (\cancel{\cos x}) = 2 \sin y - y^2 \sin x$$

partially with respect to  $y$ :

$$\frac{\partial f}{\partial y} = 2x \cos y + 2y \cos x = 2x \cos y + 2y \cos x.$$

8a Evaluate  $I = \int_1^2 \int_0^3 x^2 y \, dx \, dy$

Solution

$$I = \int_1^2 \left[ \int_0^3 x^2 y \, dx \right] dy \quad \text{picking } dx$$

$$= \int_0^3 x^2 y \, dx \Rightarrow \frac{x^{2+1}}{2+1} y = \frac{x^3}{3} y \Big|_0^3 = \left[ \frac{3^3 y}{3} \right] - [0] = \frac{9y}{3} = 3y$$

with respect to  $y$ .

$$\begin{aligned} \int_1^2 (3y) \, dy &\Rightarrow \frac{3y^{1+1}}{1+1} \Big|_1^2 = \frac{3y^2}{2} = \left[ \frac{3(2)^2}{2} - \frac{3(0)^2}{2} \right] \\ &= \frac{12}{2} - 0 = 6 \end{aligned}$$

$$\therefore I = \underline{6}.$$

8b Evaluate  $\int_C y \, dx + z \, dy + x \, dz$  where  $C$  is line from  $(2, 0, 0)$  to  $(3, 4, 5)$ .

Solution

$$I = \int_C y \, dx + z \, dy + x \, dz \quad \text{at } (x_1, y_1, z_1) \text{ to } (x_2, y_2, z_2)$$

where  $x: x_1 + t(x_2 - x_1)$

$$\text{Equating: } x = 2 + t(3 - 2) = 2 + t(1) = 2 + t \quad \frac{dx}{dt} = 1 \quad dx = dt$$

$$y = 0 + t(4 - 0) = 4t$$

$$\frac{dy}{dt} = 4 \quad dy = 4dt$$

$$z = 0 + t(5 - 0) = 5t$$

$$\frac{dz}{dt} = 5 \quad dz = 5dt.$$

Substituted on  $\int_C (y dx + z dy + x dz)$ .

$$= \int_C (4t dt + 5t (4dt) + 2+t(5dt))$$

$$= \int_C (4t dt + 20t dt + 10dt + 5t dt)$$

$$\text{collecting like terms} = \int_C (29t dt + 10dt) = \int_C (29t + 10) dt.$$

getting lower limit from  $x_1$  at 2.

$$\text{where } x = 2+t \quad \therefore 2 = 2+t \quad t = 2-2 = 0$$

getting upper limit from  $x_2$  at 3

$$\text{where } x = 2+t \quad \therefore 3 = 2+t \quad t = 3-2 = 1.$$

$$\therefore \int_0^1 (29t + 10) dt.$$

$$= \left[ \frac{29t^{1+1}}{1+1} + \frac{10t^{0+1}}{0+1} \right]_0^1 = \left[ \frac{29t^2}{2} + 10t \right]_0^1 = \left[ \frac{29(1)^2}{2} + 10(1) \right] - 0$$

$$= \frac{29}{2} + 10 = \frac{49}{2}$$

Qa Evaluate  $\int_C xy^4 ds$  over curve  $C$ , where  $C$  is the right half of circle  $x^2 + y^2 = 16$ .

Qb Integrate  $8\cos(5x+3)$ ,  $x^2 - 4x^3 + 2x^4$

Solution -

$$\bullet 8\cos(5x+3) \text{ let } 5x+3 \text{ be } u = \int 8\cos(5x+3) = \int (8\cos u)$$

$$\frac{du}{dx} = 5 \quad du = 5dx \quad dx = \frac{du}{5}$$

$$= \int 8\cos u \cdot \frac{du}{5} = \frac{8}{5} \int \cos u du$$

$$= \frac{8}{5} \sin u + C$$

$$\bullet x^2 - 4x^3 + 2x^4 \quad \int (x^2 - 4x^3 + 2x^4)$$

$$= \frac{x^{2+1}}{2+1} - \frac{4x^{3+1}}{3+1} + \frac{2x^{4+1}}{4+1} = \frac{x^3}{3} - x^4 + \frac{2x^5}{5} + C$$

$$= \frac{8}{5} \sin(5x+3) + C$$



11b Given that  $f(x, y) = x^2 - 2y^2$ , Obtain  $\frac{\partial f}{\partial y}$  and  $\frac{\partial f}{\partial x}$  from 1st principle.

Solution.

$$f(x, y) = x^2 - 2y^2$$

$$\text{first principle} = \lim_{\Delta a \rightarrow 0} \frac{f(a + \Delta a, b) - f(a, b)}{\Delta a}$$

$$\lim_{\Delta b \rightarrow 0} \frac{f(a, b + \Delta b) - f(a, b)}{\Delta b}$$

$$\therefore f(x, y) = f(x) = \lim_{\Delta x \rightarrow 0} \frac{((x + \Delta x)^2 - 2y^2) - (x^2 - 2y^2)}{\Delta x}$$

$$\text{where } (x + \Delta x)^2 =$$

$$x^2 + 2x\Delta x + \Delta x^2$$

$$\therefore \lim_{\Delta x \rightarrow 0} \frac{(x^2 + 2x\Delta x + \Delta x^2 - 2y^2) - (x^2 - 2y^2)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{2x\Delta x + \Delta x^2}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} 2x + \Delta x$$

$$= 2x + 0$$

$$\therefore f(y) = \lim_{\Delta y \rightarrow 0} \frac{(2(y + \Delta y)^2 + x^2 - x^2 - 2y^2)}{\Delta y}$$

$$= 2y$$

$$\text{where } (y + \Delta y)^2 = y^2 + 2y\Delta y + \Delta y^2$$

$$= \lim_{\Delta y \rightarrow 0} \frac{-2(y^2 + 2y\Delta y + \Delta y^2) + x^2 - x^2 - 2y^2}{\Delta y} = \frac{-2y^2 - 4y\Delta y - 2\Delta y^2 + x^2 - x^2 - 2y^2}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{-4y\Delta y - 2\Delta y^2}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} -4y - 2\Delta y$$

$$= -4y - 2(0)$$

(13a) Evaluate  $I = \int_1^3 \int_{-1}^1 \int_0^2 (x + 2y - z) dx dy dz$

$$= 4y$$

Solution.

$$I = \int_1^3 \int_{-1}^1 \int_0^2 (x + 2y - z) dx dy dz \quad \text{taking the } dx.$$

$$= \int_0^2 (x + 2y - z) dx = \left[ \frac{x^{0+1}}{0+1} + 2yx^{0+1} - \frac{zx^{0+1}}{0+1} \right]_0^2 = \frac{x^2}{2} + 2xy - xz$$

$$= \left. \frac{x^2}{2} + 2xy - xz \right|_0^2 = \left[ \frac{2^2}{2} + 2(2)y - 2(z) \right] - 0$$

$$= \left[ \frac{4}{2} - 4y - 2z \right]$$

Input back to I.

$$I = \int_1^3 \left[ \int_{-1}^1 (2 - 4y - 2z) dy \right] dz$$

Integrating with respect to y.

$$= \int_{-1}^1 (2 - 4y - 2z) dy = \left. \frac{2y^{0+1}}{0+1} - \frac{4y^{1+1}}{1+1} - \frac{2zy^{0+1}}{0+1} \right|_{-1}^1$$

$$= \left. 2y - \frac{4y^2}{2} - 2yz \right|_{-1}^1 = \left. 2y - 2y^2 - 2yz \right|_{-1}^1 = \left[ 2(1) - 2(1)^2 - 2(1)z \right] - \left[ 2(-1) - 2(-1)^2 - 2(-1)z \right]$$

$$= [2 - 2 - 2z] - [-2 + 2 + 2z] = -2z + 2 - 2 - 2z = -4z$$

inputting back to I

$$I = \int_1^3 (-4z) dz \quad \text{integrating,}$$

$$\left. \frac{-4z^{0+1}}{0+1} \right|_1^3 = \left. \frac{-4z^2}{2} \right|_1^3 = \left. -2z^2 \right|_1^3$$

$$\therefore \left[ -2(3)^2 \right] - \left[ -2(1)^2 \right]$$

$$= -18 - [-2] = -18 + 2 = -16$$

13b Find the derivatives of i)  $e^{4x}$  ii)  $x^2 e^x$  iii)  $\sin 3x$

Solution

i)  $e^{4x}$  let  $4x$  be  $u$   $\frac{du}{dx} = 4$   $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$   $\frac{dy}{du} = e^u$

$$\frac{dy}{dx} = e^u \cdot 4 = \underline{4e^{4x}}$$

ii)  $x^2 e^x$  let  $x^2$  be  $u$  let  $e^x$  be  $v$  using product rule  $u \frac{dv}{dx} + v \frac{du}{dx}$



$$\frac{du}{dx} = 2x \quad \frac{dv}{dx} = e^x$$

$$\frac{dy}{dx} = x^2(e^x) + e^x(2x)$$

$$= x^2 e^x + 2x e^x = e^x(x^2 + 2x)$$

iii  $\sin 3x$  let  $3x$  be  $u$   $\frac{du}{dx} = 3$   $\frac{dy}{du} = \cos u$

$$\therefore \frac{dy}{dx} = \frac{du}{dx} \times \frac{dy}{du} = \cos u \cdot 3$$

$$= 3 \cos 3x$$